Online Resource 1

Code Geometry, Enforcement Activation, and Exit Closure in Codified Moral Systems: An Agent-Based Model

This supplement documents the parameterization of the simulation model, the robustness of the outcome classification, the continuous surfaces that underlie the discretized results, the privilege ablation, the endogenous exit-capacity module, and the computational provenance of every reported quantity. Section S1 is taken from the frozen model sources. Sections S2 through S5 report quantities computed from the committed run outputs. Section S6 states where those outputs and the code that produced them reside.

1 Parameter-to-observable mapping

This section supplies the detailed measurement material moved from the main text. The mapping proceeds from model parameters to institutional indicators. Legibility is the extent to which compliance is classifiable through visible acts, speech, dress, attendance, dietary practice, or other public markers. Substitutability asks whether visible compliance is accepted as sufficient, or near-sufficient, for moral standing. Enforcement affordance is the ease with which deviations are detected, coded, and linked to sanctions. Centralization is the concentration of interpretive and punitive authority. Exit cost is approximated by legal penalties, kinship endogamy, economic dependence, identity foreclosure, and the social cost of disaffiliation.

Enforcement reward is observed in promotion, prestige, institutional access, or material compensation tied to disciplinary activity. Delegation is observed in specialized offices, enforcement jurisdictions, and recognized roles for monitoring and sanctioning. These indicators locate a bounded regime in a common structural space; they do not produce a ranked taxonomy of traditions.

Table 1: Code geometry dimensions and observable indicators.

Dimension	Model parameter	Observable indicators
Legibility	σ	Visible prescriptions, dress, attendance, dietary markers
Substitutability	Collapsed into σ	Visible compliance accepted as sufficient for standing
Enforcement affordance	π, κ	Violation categories, complaint channels, sanction pathways
Centralization	Cadre and monopoly	Adjudicating bodies, orthodoxy monopoly, restricted appeals
Exit capacity	base opportunity, δ	Legal, kinship, economic, identity, and social outside options

Table 2: Main-text model parameters and observable interpretation.

Parameter	Symbol	Reported value	Interpretation
Legibility weight	σ	swept	Weight of external compliance
Enforcement reward	π	swept	Return to successful punishment
Enforcement cost	κ	0.08	Backlash risk per act
Enforcer quota	q	0.08	Fraction designated as cadre
Alt. degradation	δ	swept	Loss of outside-option quality
Drift speed	η	swept	Rate of endogenous degradation

No single dimension is sufficient to determine regime type. High legibility with open exit can produce active but leaky enforcement; high exit cost without activation conditions can retain members without producing an apparatus. The same tradition may occupy different locations across periods, states, or rival institutions. The mapping compares configurations and does not measure latent interior states.

2 Model parameters

Two frozen model sources generate the reported results. The main experiments (activation boundary, imposed-closure dose-response, shock factorial, closure-decoupling, and privilege ablation) use `src/religion_fundamentalism_abm_v2_7.py`. The endogenous exit-capacity experiments of Section S5 use `src/religion_fundamentalism_abm_v3_3.py`, which reproduces the former exactly when `exit_capacity_mode` is `off`.

Tables S1.1 through S1.8 list every parameter of the main model with the default value declared in the frozen source and the role that value plays. The rightmost column records how the reported experiments set the parameter: *default* means the source default is used, *swept* means the experimental design varies it across the values given in the Methods, and *set* means the sweep launcher passes an explicit value that differs from the source default. Table S1.9 lists the additional parameters introduced by the endogenous exit-capacity module. Table S1.10 gives the calibration parameter set that the sweep launcher supplies, which is provided in the repository at `results/BEST_PARAMS_frozen.json`.

Table 3: Table S1.1: Population, network, and schedule.

Parameter	Default	Role	In reported runs
`n`	300	Number of agents at initialization	set to 350
`graph`	scale_free	Network generator; Barabási–Albert with attachment parameter $m = \max\{2, \text{round}[\log(N+1)]\}$	default
`seed`	1	Seed for network generation, trait draws, and all stochastic updates	swept, 1 to 30
`steps`	350	Number of simulated steps	set to 450
`fixed_y0`	False	When true, freezes the orthodox belief position at its initial value	default

Note on launcher-supplied values. The sweep launcher `scripts/run_v2_7_endogenous_delta_sweep.py` reads seven parameters from a calibration file supplied through its `--best`

Table 4: Table S1.2: Code geometry and observation.

Parameter	Default	Role	In reported runs
`sigma`	0.75	Weight of externally legible compliance relative to intrinsic cultivation in the perceived-morality signal	swept
`v_obs`	0.90	Probability that a compliance marker is observed	default
`a_obs`	0.05	Probability that intrinsic cultivation is observed	default
`h_obs`	0.15	Probability that an agent's belief position is observed for the heresy-distance term	default

argument: `exit_cost`, `exit_block_exponent`, `exit_commit_steps`, `exit_opportunity_threat_coeff`, `membership_benefit`, `membership_benefit_threat`, and `shock_strength`. Its `load_best_params` function reads these seven keys and its `run_one` function passes each of them to the model as a command-line flag. For all seven, the value that ran differs from the frozen source default, so each is a *set* row above rather than a *default* row.

The calibration file is provided in the repository at `results/BEST_PARAMS_frozen.json`. Table S1.10 gives its contents.

An eighth key, `exit_opportunity_base`, is present in the calibration file but is not read by `load_best_params`. The launcher passes its own `--base-opp`, which is 0.30 in every reported sweep and is recorded as `base_opp=0.3` in each `sweep_report.txt`.

How the seven values were confirmed. The committed run outputs do not record a command line, so each value was recovered from what the runs wrote and then checked by replay. Three methods agree.

First, algebraic inversion. `agent_summary.csv` records `exit_cost_eff_last`, which the model computes as `exit_cost` $(1+0.03d_i)+0.25s_i$ with $s_i \in \{0,1\}$. Inverting this over every agent with a nonzero entry returns exactly 1.0 and never 0.40.

Second, direct reading of echoed columns. In `recon/factorial_shock_on`, the `threat` column steps from 0 to exactly 0.1 across the first scheduled shock and peaks at 0.1050330,

Table 5: Table S1.3: Agent behaviour, learning, and social dynamics.

Parameter	Default	Role	In reported runs
c_epc	0.12	Utility cost of displaying the visible compliance marker	default
eps_cult	0.06	Utility cost of investing in intrinsic cultivation	default
eta_cult	0.07	Utility return to intrinsic cultivation	default
beta_epc	5.0	Logistic slope converting net status gain into the probability of displaying compliance	default
beta_pun	6.0	Logistic slope converting the perceived compliance deficit into punishment probability	default
k_h	10.0	Logistic slope of the heresy-distance term	default
rho_r	0.04	Step size for updating enforcement propensity after a punishment outcome	default
rho_b	0.03	Step size for updating rigidity after a punishment outcome	default
w_m	1.0	Weight on the perceived-morality shortfall in the punishment deficit	default
w_heresy	0.6	Weight on the heresy-distance term in the punishment deficit	default
d0	0.18	Baseline heresy-distance threshold above which belief counts as deviant	default
norm_weight	0.55	Weight on the local compliance norm in the status gain from displaying compliance	default
shame_weight	0.70	Weight on the local enforcement norm in that status gain	default
noise_x	0.02	Standard deviation of per-step noise on intrinsic cultivation	default
cult_step	0.04	Maximum per-step change in intrinsic cultivation	default
F_star	0.45	Threshold on the fundamentalism index used to report prevalence	default
L_alpha	1.5	First shape parameter of the Beta distribution of literalism	default
L_beta	4.5	Second shape parameter of that distribution	default
theta_L_gain	0.12	Sensitivity of an agent's effective strictness threshold to its literalism	default
d0_L_gain	0.08	Sensitivity of an agent's effective heresy threshold to its literalism	default

Table 6: Table S1.4: Policing economy.

Parameter	Default	Role	In reported runs
`pi_reward`	0.22	Coefficient on the legibility index in the institutional return to a successful punishment	swept
`kappa_cost`	0.08	Base utility cost of issuing a punishment	default
`lam_punish`	0.25	Utility penalty imposed on the punished agent	default
`backlash_base`	0.25	Baseline probability that a punishment provokes backlash against the punisher	default
`backlash_cost`	0.30	Utility cost borne by the punisher when backlash occurs	default
`backlash_sensitivity`	1.0	Sensitivity of backlash probability to the illegibility of the code	default
`punish_floor`	0.08	Minimum step-level fraction of active agents punished that gates any update of outside-option closure	set to 0.08

the value implied by an increment of 0.1 under the 0.97 per-step relaxation. Regressing the `stay_value` column on `threat` in the same run returns an intercept of 0.0843663 and a slope of 0.1055400, which give `membership_benefit` = 0.0633663 and `membership_benefit_threat` = 0.1055400 once the legibility term is subtracted. The agent-level `disengaged` flag is set at ⌊`exit_commit_steps`/2⌋; its observed threshold of 6 implies 12 and excludes 8.

Third, replay, which is decisive. Re-running `src/religion_fundamentalism_abm_v2_7.py` at the recorded cell and seed reproduces the committed `metrics.csv` and `agent_summary.csv` exactly under the parameter set in Table S1.10, for one run drawn from each of `recon/exogenous_delta_fixed`, `recon/boundary_sealed`, and `recon/factorial_shock_on`. Substituting the frozen-source defaults breaks the reproduction in all three: terminal exit moves from 0.046 to 0.871 in the first run and from 0.003 to 0.354 in the second.

Identifiability caveat. The three parameters `shock_strength`, `exit_opportunity_threat_coeff`, and `membership_benefit_threat` all enter the model multiplied by threat. Every

Table 7: Table S1.5: Exit desire, opportunity, and cost.

Parameter	Default	Role	In reported runs
`enable_exit`	True	Master switch for the exit process	default
`exit_threshold`	-1.0	Utility level below which exit desire begins to accumulate	set to -1.0
`exit_prob_slope`	3.0	Logistic slope converting exit drive into exit desire	default
`exit_cost`	0.40	Base cost of leaving, before degree and lock-in adjustments	set to 1.00
`exit_degree_coeff`	0.03	Rate at which network degree raises the effective exit cost	default
`exit_epc_lockin_coeff`	0.25	Rate at which an agent's accumulated visible compliance raises its exit cost	default
`exit_min_cost`	0.0	Floor on the effective exit cost	default
`exit_opportunity_base`	0.6	Baseline logit intercept of the exit-opportunity draw	set to 0.30
`exit_opportunity_deg_coeff`	0.03	Rate at which degree lowers exit opportunity	default
`exit_opportunity_threat_coeff`	1.5	Rate at which external threat lowers exit opportunity	set to 2.9081
`exit_block_exponent`	2.5	Exponent applied to the raw opportunity, concentrating opportunity at high values	set to 5.9806
`exit_block_floor`	0.02	Additive floor on exit opportunity after the exponent is applied	default
`exit_commit_steps`	8	Consecutive steps for which desire must persist before an opportunity draw is taken	set to 12
`exit_cooldown`	0	Steps an agent must wait after a failed exit attempt	default
`exit_rewire_fraction`	0.90	Fraction of an exiting agent's incident edges removed on departure	default
`alpha_punish_revalue`	0.0	Weight by which received punishment revalues staying inside the group	set to 0.00
`mu_membership_reward`	0.0	Weight on the membership reward that offsets exit desire	set to 0.00
`membership_benefit`	0.03	Base per-step benefit of remaining a member	set to 0.0634
`membership_benefit_sigma`	0.08	Sensitivity of that benefit to code legibility	default
`membership_benefit_threat`	0.10	Sensitivity of that benefit to external threat	set to 0.1055

Table 8: Table S1.6: Outside-option closure.

Parameter	Default	Role	In reported runs
`delta_outside_degrade`	0.0	Imposed baseline degradation of the outside option, denoted δ_0	swept
`eta_delta_drift`	0.0	Rate at which closure drifts toward its target, denoted η	swept
`delta_mode`	legacy	Selects the closure specification: `legacy` makes the target a function of enforcer share, `decoupled` makes it a function of active-agent punishment incidence, and `exogenous` holds closure fixed	swept
`delta_cap`	0.85	Upper bound on the closure target in decoupled mode, denoted $\delta_{\max}$	default
`delta_kappa`	3.0	Coupling strength from punishment incidence to closure in decoupled mode	swept over 1.5, 3.0, 6.0

sweep except `recon/factorial_shock_on` runs with an empty shock schedule, so threat is identically zero and those three parameters are inert and formally unidentifiable there. This was confirmed directly: replaying the `recon/exogenous_delta_fixed` run with those three set to the source defaults still reproduces the committed output bit-for-bit. They are identified only in `recon/factorial_shock_on`, where setting `exit_opportunity_threat_coeff` to 1.5 breaks bit-for-bit identity and 2.908096432532684 restores it.

Table 9: Table S1.7: Institutional privilege bundle.

Parameter	Default	Role	In reported runs
`enforcer_quota_frac`	0.08	Target cadre size as a fraction of active agents	default
`cap_to_enforcer`	0.25	Institutional capital required for first-round cadre eligibility	default
`A_enforcer_monopoly`	0.35	Controller authority above which the punishment monopoly excludes non-enforcers	default
`enforcer_punish_mult`	1.5	Multiplier on an enforcer's punishment probability	default
`non_enforcer_punish_eps`	0.02	Operative multiplier applied to non-enforcers when the monopoly is off	default
`non_enforcer_punish_mult`	0.25	Declared non-enforcer multiplier; passed by the sweeps but not used in the operative branch	default
`kappa_cap_discount`	0.20	Fraction by which accumulated capital reduces the base punishment cost	default
`enforcer_kappa_mult`	0.30	Further multiplier on an enforcer's punishment cost, a 70 percent role discount	default
`enforcer_backlash_mult`	0.25	Multiplier on an enforcer's backlash probability, 75 percent protection	default
`cap_gain_per_punish`	0.15	Institutional capital added per successful punishment	default
`cap_max`	2.0	Ceiling on accumulated institutional capital	default
`cap_decay`	0.005	Per-step decay rate of institutional capital	default
`service_decay`	0.02	Per-step decay rate of accumulated service	default
`budget_base`	0.15	Constant term of the controller patronage budget	default
`budget_threat_gain`	0.60	Coefficient on threat in the patronage budget	default

Table 10: Table S1.8: Institutional controller and threat schedule.

Parameter	Default	Role	In reported runs
`shock_schedule`	100, 220, 320	Steps at which external threat is raised	swept: empty, or 100, 220, 320
`shock_strength`	0.25	Increment added to threat at each scheduled shock	set to 0.10
`relax_rate`	0.03	Per-step proportional decay of threat between shocks	default
`A_gain_threat`	2.0	Rate at which threat raises controller authority through $A_t = 1 - \exp(-A_{\text{gain}} T_t)$	default
`tighten_gain_pi`	0.20	Increase in enforcement reward per unit threat	default
`tighten_gain_lam`	0.20	Increase in punishment severity per unit threat	default
`tighten_gain_d0`	0.08	Reduction in the heresy threshold per unit threat	default
`baseline_pi`	-1.0	Sentinel; a negative value makes the run adopt `pi_reward` as the baseline	default
`baseline_lam`	-1.0	Sentinel for the punishment-severity baseline	default
`baseline_d0`	-1.0	Sentinel for the heresy-threshold baseline	default

Table 11: Table S1.10: The calibration parameter set, in `results/BEST_PARAMS_frozen.json`.

Parameter	Source default	Value that ran	How it was confirmed
`exit_cost`	0.40	1.0	algebraic inversion; replay
`exit_block_exponent`	2.5	5.980606901200484	replay
`exit_commit_steps`	8	12	disengagement threshold; replay
`exit_opportunity_threat_coeff`	1.5	2.908096432532684	replay, shock-on only
`membership_benefit`	0.03	0.0633663491490371[illegible]	`stay_value` column; replay
`membership_benefit_threat`	0.10	0.10553999169992[illegible]	`stay_value` regression, shock-on only
`shock_strength`	0.25	0.1	`threat` column, shock-on only

Table 12: Table S1.9: Additional parameters of the endogenous exit-capacity module.

Parameter	Default	Role	In reported runs
`exit_capacity_mode`	off	Selects the capacity specification; `off` reproduces the main model exactly, `endogenous` replaces global closure with a per-agent capacity	set to endogenous
`ec_init_random`	True	Draws per-agent outside ties and economic independence from Beta(2,2) at initialization	default
`ec_outside_ties_init`	1.0	Initial outside-tie level when homogeneous initialization is selected	default
`ec_tie_decay`	0.01	Per-step decay rate of outside social ties	default
`ec_tie_renewal_base`	0.006	Constant term of outside-tie renewal; the tie channel closes when crowd-out exceeds this value	default
`ec_tie_renewal`	0.004	Renewal of outside ties proportional to current ties	default
`ec_tenure_crowdout`	0.002	Rate at which tenure inside the group crowds out outside ties	swept over 0.0005, 0.002, 0.008
`ec_econ_recovery_base`	0.004	Constant term of economic-independence recovery	default
`ec_econ_recovery`	0.002	Recovery of economic independence proportional to its current level	default
`ec_dependence_rate`	0.01	Rate at which membership reward erodes economic independence	default
`ec_hetero_sd`	0.25	Standard deviation of per-agent heterogeneity in capacity	default
`lambda_exit_opportunity`	1.0	Weight of capacity on the exit-opportunity channel	default
`lambda_exit_willingness`	1.0	Weight of capacity on the exit-willingness channel	default
`turnover_mode`	off	Enables cohort turnover through death and replacement	set to off
`death_hazard`	0.002	Per-step death hazard when turnover is on	default
`born_inside_frac`	0.8	Fraction of the population initialized as born inside the group	swept over 0.0 and 0.8
`emit_panel`	False	Emits the per-step per-agent panel used for the anti-circularity test	set to on for the panel runs

3 Classification robustness

The canonical classifier defines capture as retention together with activation: a run is retained when `final_exit_rate` ≤ 0.20 and active when `max_punish` ≥ 0.10. This section reports the sensitivity of that definition to the activation threshold, the vacuity of the concentration criterion that the definition omits, and the relationship between COLLAPSE and activation.

3.1 Activation-threshold sensitivity

Table S2.1 varies the activation threshold and records, for each value, the capture rate in the imposed-closure experiment, the capture rate at the sealed boundary, whether the imposed-closure dose-response remains monotone, and whether the low-observability, low-reward cell at $\sigma = 0.25$, $\pi = 0.05$ remains inactive.

Table 13: Table S2.1: Capture rate and qualitative behaviour against the activation threshold.

Threshold	Imposed-closure capture	Sealed-boundary capture	Dose-response	$\sigma = 0.25$, $\pi = 0.05$ cell
0.05	0.253	0.816	monotone	breaks, 90 of 180 capture
0.075	0.198	0.694	monotone	holds, 0 of 180
0.10 (canonical)	0.198	0.661	monotone	holds, 0 of 180
0.15	0.186	0.473	monotone	holds, 0 of 180
0.20	0.006	0.070	degenerates	holds, 0 of 180

Both the dose-response and the necessity result survive across the band $[0.075, 0.15]$, and the canonical value 0.10 lies in the interior of that band. The two edges fail in opposite directions. At the loose edge, 0.05, the $\sigma = 0.25$, $\pi = 0.05$ cell ceases to be inactive and returns 90 of 180 runs as capture, so the necessity result no longer holds. At the strict edge, 0.20, capture falls to approximately zero everywhere, with an imposed-closure capture rate of 0.006 and a sealed-boundary rate of 0.070, so the dose-response degenerates and carries no information. The reported band is therefore joint: it is the set of thresholds at which both results hold simultaneously. Sources: `recon/exogenous_delta_fixed` and `recon/boundary_sealed`.

3.2 Vacuity of the concentration criterion

An earlier form of the classifier gated capture on an enforcer punishment share of at least 0.70. Reclassifying every committed run with and without that gate changes 0 of 9,570 classifications. The 9,570 runs comprise 4,050 from the shock-off factorial, 1,620 from the imposed-closure experiment, 1,890 from the sealed boundary, 1,890 from the open boundary, and 120 from the ablation ceiling.

The reason the gate is inert is distributional. Enforcer share has no mass near zero in any of the five datasets: the fraction of runs at or below 0.05 is 0.000 throughout, and the median lies between 0.97 and 0.98. The statistic saturates high wherever punishment occurs at all, which makes it redundant with activation rather than a graded measure of concentration. It is therefore reported descriptively and excluded from the capture definition.

3.3 COLLAPSE and activation

Every COLLAPSE run in the committed data falls below the canonical activation threshold. Table S2.2 gives the count and the observed range of maximum punishment incidence by dataset.

Table 14: Table S2.2: COLLAPSE runs and their maximum punishment incidence.

Dataset	COLLAPSE runs	Of which `max_punish` < 0.10	min / median / max
`results/v2.5_methodology_paper_canonical`	45	45 (100 percent)	0.037 / 0.063 / 0.094
`results/v2.5_corrected_three_regime_confirm`	45	45 (100 percent)	0.037 / 0.063 / 0.094
`results/v2.5_corrected_sweep_regime_search_fast`	18	18 (100 percent)	0.037 / 0.059 / 0.094
All `recon/` sweeps	0	not applicable	not applicable

All 108 COLLAPSE runs are inactive, and the largest maximum punishment incidence observed among them is 0.094, below the 0.10 activation threshold. Depopulation in these runs is therefore driven by open exit rather than by enforcement, and COLLAPSE is properly described as depopulation without activation.

This has a direct consequence for the structure of the classifier. Reclassifying all committed data as activation crossed with retention, with no separate COLLAPSE rule, changes 108 of 28,581 runs, or 0.38 percent, and every change is COLLAPSE to QUIET. The inactive cell with exit strictly between 0.20 and 0.90 contains 560 runs, or 1.96 percent, all already labelled QUIET. COLLAPSE is retained only as a descriptive flag for the exit ≥ 0.90 tail within the inactive region, not as a separate dynamical regime.

4 Continuous surfaces

The classification in Section S2 discretizes two continuous quantities. This section reports those quantities directly, so that the reported results can be read without reference to any threshold. Each entry is a mean over 30 seeds for the corresponding cell.

4.1 Open exit

The open-exit condition fixes $\delta_0 = 0$ and comprises 1,890 runs on a grid of 9 values of σ by 7 values of π by 30 seeds. Source: `recon/boundary_open/sweep_seed_results.csv`.

Table 15: Table S3.1: Mean terminal exit rate, open exit.

$\sigma \backslash \pi$	0.01	0.03	0.05	0.10	0.15	0.25	0.50
0.05	0.176	0.173	0.176	0.174	0.172	0.542	0.500
0.15	0.096	0.097	0.094	0.374	0.490	0.464	0.440
0.25	0.046	0.044	0.044	0.446	0.433	0.405	0.426
0.35	0.016	0.016	0.319	0.409	0.383	0.403	0.446
0.45	0.005	0.005	0.400	0.385	0.382	0.400	0.448
0.55	0.002	0.219	0.380	0.384	0.393	0.417	0.440
0.65	0.000	0.306	0.368	0.378	0.384	0.424	0.419
0.75	0.000	0.343	0.366	0.374	0.389	0.400	0.399
0.95	0.000	0.319	0.318	0.324	0.322	0.317	0.310

Table 16: Table S3.2: Mean maximum punishment incidence, open exit.

$\sigma \backslash \pi$	0.01	0.03	0.05	0.10	0.15	0.25	0.50
0.05	0.050	0.050	0.050	0.050	0.050	0.091	0.110
0.15	0.050	0.050	0.050	0.064	0.098	0.109	0.148
0.25	0.051	0.051	0.051	0.102	0.110	0.132	0.172
0.35	0.046	0.047	0.074	0.117	0.130	0.156	0.181
0.45	0.046	0.046	0.105	0.133	0.152	0.173	0.181
0.55	0.044	0.066	0.126	0.149	0.164	0.173	0.180
0.65	0.042	0.093	0.136	0.156	0.163	0.171	0.174
0.75	0.039	0.118	0.143	0.154	0.162	0.162	0.163
0.95	0.033	0.128	0.130	0.136	0.138	0.136	0.135

4.2 Sealed exit

The sealed-exit condition fixes $\delta_0 = 0.95$ on the same grid, again 1,890 runs. Source: `recon/boundary_sealed/sweep_seed_results.csv`.

Table 17: Table S3.3: Mean terminal exit rate, sealed exit.

$\sigma\backslash\pi$	0.01	0.03	0.05	0.10	0.15	0.25	0.50
0.05	0.008	0.008	0.009	0.007	0.008	0.084	0.107
0.15	0.004	0.004	0.004	0.047	0.080	0.099	0.138
0.25	0.002	0.002	0.002	0.077	0.081	0.105	0.141
0.35	0.001	0.001	0.042	0.076	0.096	0.128	0.128
0.45	0.000	0.000	0.068	0.089	0.110	0.117	0.118
0.55	0.000	0.023	0.068	0.100	0.108	0.113	0.110
0.65	0.000	0.041	0.073	0.092	0.100	0.097	0.097
0.75	0.000	0.058	0.076	0.081	0.083	0.082	0.083
0.95	0.000	0.056	0.053	0.055	0.055	0.054	0.054

Table 18: Table S3.4: Mean maximum punishment incidence, sealed exit.

$\sigma\backslash\pi$	0.01	0.03	0.05	0.10	0.15	0.25	0.50
0.05	0.050	0.050	0.050	0.050	0.050	0.107	0.126
0.15	0.050	0.050	0.050	0.087	0.113	0.131	0.186
0.25	0.051	0.051	0.051	0.121	0.129	0.164	0.196
0.35	0.046	0.047	0.099	0.136	0.159	0.192	0.202
0.45	0.046	0.046	0.130	0.163	0.183	0.197	0.204
0.55	0.044	0.086	0.147	0.182	0.192	0.200	0.202
0.65	0.042	0.125	0.161	0.179	0.189	0.197	0.195
0.75	0.039	0.151	0.173	0.179	0.189	0.189	0.189
0.95	0.033	0.154	0.158	0.164	0.166	0.165	0.165

Comparing Tables S3.1 and S3.3 shows that sealing the exit lowers terminal exit throughout the grid. Comparing Tables S3.2 and S3.4 shows that it also raises enforcement intensity across most of the grid, which is the continuous form of the coupling between the two axes at the frontier margin.

4.3 Imposed closure

The imposed-closure experiment holds closure fixed at a series of levels and comprises 1,620 runs over 3 values of σ by 3 values of π by 6 values of δ_0 by 30 seeds. Source: `recon/`

`exogenous_delta_fixed/sweep_seed_results.csv`.

Table 19: Table S3.5: Mean maximum punishment and mean terminal exit by imposed closure, pooled over σ and π.

δ_0	Mean maximum punishment	Mean terminal exit	n
0.00	0.136	0.332	270
0.20	0.139	0.324	270
0.40	0.143	0.313	270
0.60	0.148	0.288	270
0.80	0.155	0.216	270
0.95	0.161	0.072	270

Table 20: Table S3.6: Mean maximum punishment over σ and π at $\delta_0 = 0.95$.

$\sigma\backslash\pi$	0.05	0.25	0.50
0.25	0.051	0.164	0.196
0.75	0.173	0.189	0.189
0.95	0.158	0.165	0.165

Raising imposed closure from 0.00 to 0.95 lowers mean terminal exit from 0.332 to 0.072 while raising mean maximum punishment only from 0.136 to 0.161. Closure acts principally on retention rather than on activation.

4.4 The necessity result without a threshold

The necessity claim has a form that requires no classification. At the low-observability, low-reward corner $\sigma = 0.25$, $\pi = 0.05$, mean maximum punishment over 30 seeds is 0.051 at $\delta_0 = 0.0$ and 0.051 at $\delta_0 = 0.95$, a change of +0.000, while mean terminal exit falls from 0.044 to 0.002. Sealing the exit in this corner removes the departures without producing any enforcement. Both values lie far below the 0.10 activation band, so the statement is a claim about the continuous enforcement surface and does not depend on where the activation threshold is placed.

5 Privilege ablation

The ablation isolates the contribution of the institutional privilege bundle to the concentration of punishment. It compares a floor arm, in which every privilege is removed, against a ceiling arm carrying the complete bundle, and then adds each privilege back individually. All metrics are computed over agents active at the end of the run, and each arm contains 120 runs. Source: `recon/privilege_ablation/ablation_seed_results.csv`.

5.1 Floor, ceiling, and the random-allocation null

The null holds the observed active-agent punishment volume fixed and reallocates those events uniformly at random across active agents, which gives the concentration that arises from counting alone.

Table 21: Table S4.1: Floor and ceiling concentration against the random-allocation null.

Arm	Top-5 percent share	Gini	Gini, random null	Terminal exit rate
Floor	0.091	0.240	0.028	0.585
Ceiling	0.580	0.901	0.062	0.331

The privilege-manufactured share of top-5 percent concentration is $(0.580-0.091)/0.580 = 0.843$, that is 84 percent. The floor Gini of 0.240 sits only modestly above its random null of 0.028, so the residual concentration in a privilege-free model is small. Concentration is a product of the privilege architecture rather than an emergent property of the underlying interaction.

5.2 Single-privilege add-back

Each privilege is added individually to the floor, holding all others removed, with 120 seeds per arm. Recovery is expressed as a fraction of the floor to ceiling span in top-5 percent share, which runs from 0.0907 to 0.5799.

Table 22: Table S4.2: Concentration after adding one privilege to the floor.

Arm	Top-5 percent share	Gini	Share of span recovered
Floor	0.0907	0.2402	not applicable
Quota	0.0907	0.2402	0.0 percent
Monopoly	0.0907	0.2402	0.0 percent
Punishment multiplier	0.2505	0.6317	32.7 percent
Backlash protection	0.0907	0.2402	0.0 percent
Capital gain	0.0907	0.2402	0.0 percent
Budget patronage	0.0907	0.2402	0.0 percent
Cost discount	0.0907	0.2402	0.0 percent
Ceiling	0.5799	0.9007	100 percent

Seven of the eight single privileges are exactly inert, reproducing the floor values to four decimal places. The exception is the punishment multiplier, which on its own recovers 32.7 percent of the span. The accurate statement is therefore conjunctivity with one exception rather than conjunctivity outright: concentration requires the privileges jointly, except for the coercive-power multiplier, which is the single load-bearing privilege. The seven inert privileges are inert for a common reason. Each acts only on a cadre, and the quota-less floor never forms one, so there is no differentiated group for them to act upon.

The multiplier exception carries a volume caveat that the concentration statistics alone do not show. Table S4.2 reports shares and Gini coefficients, which are scale-free and can be computed over any nonzero punishment count. The multiplier arm issues a mean of 393.2 punishments per run, against 75,611.8 at the floor and 19,100.6 at the ceiling, a volume roughly two orders of magnitude below either reference arm. It also retains almost the entire population, with a mean terminal exit rate of 0.010 against 0.585 at the floor and 0.331 at the ceiling, so the 32.7 percent recovery is a concentration measured over about 1.1 punishment events per surviving active agent across 450 steps. The exception is real but thin: it establishes that the coercive-power multiplier is the one privilege that can concentrate sanctioning without a cadre, not that it reproduces the enforcement volume of the full bundle.

5.3 Concentration does not discriminate capture

Concentration is high wherever punishment occurs and therefore does not separate the outcome classes. Table S4.3 reports active-agent concentration by recomputed regime in the imposed-closure experiment.

Table 23: Table S4.3: Concentration and exit by regime, imposed-closure experiment.

Regime	n	Top-5 percent share	Gini	Terminal exit rate
CAPTURE	320	0.678	0.918	0.105
MIXED	1120	0.625	0.903	0.339
QUIET	180	0.425	0.713	0.021

Among active systems, that is CAPTURE against MIXED, the two concentration measures are nearly identical, 0.678 against 0.625 for top-5 percent share and 0.918 against 0.903 for the Gini, while terminal exit differs by roughly a factor of three, 0.105 against 0.339. Retention rather than concentration separates the regimes. QUIET sits lower on both concentration measures only because it is barely active, which is an activation effect rather than a retention one, and is the reason the non-capture classes must not be pooled.

6 Endogenous exit capacity

The endogenous exit-capacity module replaces the global closure parameter with a per-agent capacity built from outside social ties and economic independence. This section reports the closure threshold of the tie channel, the generational arms, and the anti-circularity test. Model: `src/religion_fundamentalism_abm_v3_3.py`. Aggregated outputs and per-run artifacts are under `results/v3_3_endogenous_capacity_v2/`.

6.1 Closure of the tie channel

The tie channel has fixed point $o^* = (\text{renewal base} - \text{crowd-out})/(\text{decay} - \text{renewal})$, and closes when $o^* \leq 0$, which occurs when crowd-out exceeds the base renewal rate of 0.006. Table S5.1 reports a sweep over crowd-out at $\mu = 0.0$ with 5 seeds per level. Source: `results/v3_3_endogenous_capacity_v2/closure_summary.csv`.

Table 24: Table S5.1: Closure sweep over tenure crowd-out.

Crowd-out	Predicted o^*	Tie channel closes	Mean final exit capacity	Exit over last 100 steps
0.0005	+0.917	no	0.9302	0.0415
0.002	+0.667	no	0.8392	0.0544
0.008	−0.333	yes	0.5166	0.0685

The behaviour is a threshold rather than a ratchet. At $\mu = 0.0$ the economic channel remains high, so total capacity floors at 0.5166 when the tie channel closes rather than falling to zero. Driving capacity to zero requires closing the economic channel as well, which needs μ above its own threshold.

6.2 Generational arms

With the economic channel inactive at $\mu = 0.0$ and 5 seeds per arm, the born-inside cohort has lower terminal capacity. Source: `results/v3_3_endogenous_capacity_v2/generational_summary.csv`.

Table 25: Table S5.2: Generational arms, economic channel inactive.

Born-inside fraction	Exit capacity at step 449	Cumulative exit
0.0	0.8854	0.5747
0.8	0.7721	0.5680

With the economic channel live, the same comparison is run at $\mu \in \{0.6, 0.8\}$, both crowd-out levels, both born-inside fractions, and 10 seeds. The economic fixed point is $e^* = 0.004/(0.01\mu)$, giving 0.667 at $\mu = 0.6$ and 0.500 at $\mu = 0.8$, both below 1, and the observed mean economic independence of 0.52 to 0.74 confirms that the channel is genuinely interior rather than clipped at its bound. Source: `results/v3_3_endogenous_capacity_v2/live_econ_summary.csv`.

Table 26: Table S5.3: Generational arms, economic channel live.

μ	Crowd-out	Born inside	Exit capacity at step 449	Cumulative exit
0.6	0.002	0.0	0.7473	0.5230
0.6	0.002	0.8	0.6015	0.5117
0.6	0.008	0.0	0.4760	0.4060
0.6	0.008	0.8	0.3542	0.2943
0.8	0.002	0.0	0.6686	0.5147
0.8	0.002	0.8	0.5398	0.4757
0.8	0.008	0.0	0.4008	0.3560
0.8	0.008	0.8	0.3006	0.2353

A born-inside fraction of 0.8 gives lower terminal capacity than a fraction of 0.0 in every parameter cell and in all 40 of 40 seed-matched comparisons. Mean cumulative exit is likewise lower in all four parameter cells; at the level of individual replicates the comparison holds in 32 of 40 cases, and every reversal or tie occurs at the lower crowd-out level of 0.002. Both structural knobs act monotonically: higher economic dependence and higher crowd-out each lower capacity.

6.3 Anti-circularity

The concern this test addresses is that exit capacity might be responding to enforcement, which would make capture follow by construction. The test proceeds in two stages over a panel of 93,268 transitions in 300 clusters. Source: `results/v3_3_endogenous_capacity_v2/anticircularity_panel.csv`.

First, the capacity update is reconstructed from its declared inputs, namely ties, economic independence, tenure, and heterogeneity. The maximum residual across all transitions is 3.331×10^{-16}, which is machine zero. The update therefore contains no undeclared term.

Second, that structural residual is regressed on lagged punishment received, lagged punishment delivered, and enforcer status. Table S5.4 reports the coefficients.

Table 27: Table S5.4: Regression of the structural residual on enforcement history.

Regressor	Coefficient	Standard error	p
Lagged punishment received	7.96×10^{-19}	[VERIFY]	0.188
Lagged punishment delivered	5.17×10^{-20}	[VERIFY]	0.604
Enforcer status	-4.83×10^{-19}	[VERIFY]	0.434

[VERIFY: standard errors for the three coefficients in Table S5.4. The source results tables record coefficients and p-values only.]

All three coefficients are of order 10^{-19} or smaller, that is at the scale of the machine-zero residual itself, and none approaches conventional significance. Enforcement has no structural effect on exit capacity, so the capacity mechanism is an independent input to capture rather than a restatement of it.

7 Computational reproducibility

7.1 Committed run outputs

Every reported quantity derives from one of the committed aggregate files listed in Table S6.1. Each row of a `sweep_seed_results.csv` file is one run, and the per-run directories beneath it hold that run's `metrics.csv`, with one row per step, and `agent_summary.csv`, with one row per agent.

The parameter set behind these sweeps is provided in the repository, in the file `results/BEST_PARAMS_frozen.json`, and is reproduced in Table S1.10.

Table 28: Table S6.1: Committed aggregate outputs.

Experiment	Path	Runs
Shock factorial, off	`recon/factorial_shock_off/sweep_seed_results.csv`	4,050
Shock factorial, on	`recon/factorial_shock_on/sweep_seed_results.csv`	4,050
Closure decoupling, $\kappa = 1.5$	`recon/decoupled_k1.5/sweep_seed_results.csv`	4,050
Closure decoupling, $\kappa = 3.0$	`recon/decoupled_k3.0/sweep_seed_results.csv`	4,050
Closure decoupling, $\kappa = 6.0$	`recon/decoupled_k6.0/sweep_seed_results.csv`	4,050
Imposed closure	`recon/exogenous_delta_fixed/sweep_seed_results.csv`	1,620
Activation boundary, open	`recon/boundary_open/sweep_seed_results.csv`	1,890
Activation boundary, sealed	`recon/boundary_sealed/sweep_seed_results.csv`	1,890
Privilege ablation	`recon/privilege_ablation/ablation_seed_results.csv`	120 per arm
Endogenous exit capacity	`results/v3_3_endogenous_capacity_v2/`	see Section S5

7.2 Drivers and aggregators

Table S6.2 names the script that produced each set of outputs and the script that reduces them. The classifier in `src/regime_classifier.py` is the single source of truth for discrete outcomes and carries a self-check.

Table 29: Table S6.2: Drivers and aggregators.

Role	Script
Main sweep driver	`scripts/run_v2_7_endogenous_delta_sweep.py`
Factorial launcher	`recon/run_factorial.sh`
Decoupling launcher	`recon/run_decoupled.sh`
Privilege ablation driver	`recon/run_privilege_ablation.py`
Endogenous capacity driver	`scripts/run_v3_3_experiments.py`
Boundary aggregator	`recon/analyze_boundary.py`
Imposed-closure aggregator	`recon/analyze_exogenous_fixed.py`
Decoupling aggregator	`recon/analyze_decoupled.py`
Endogenous capacity aggregator	`scripts/aggregate_v3_3.py`
Outcome classifier	`src/regime_classifier.py`

7.3 Per-run configuration records

Each run of the endogenous exit-capacity experiments writes a `run_config.json` into its own output directory. That file records the full command line, the resolved model parameters, the creation and completion timestamps, the model file path, and the SHA-256 digest of the model source under the key `code_sha256`. A reader can therefore confirm that a given committed run was produced by the exact source file now in the repository.

The main sweeps under `recon/` do not write a per-run configuration record. Their launch parameters are recorded in three places. The swept parameters appear as columns of `sweep_seed_results.csv` and as the directory names beneath it. The population-level settings `base_opp`, `punish_floor`, `exit_threshold`, and `capture_exit_cap` appear in the `sweep_report.txt` written alongside it. The seven calibration-supplied parameters appear in neither, and are instead provided in the repository as the file `results/BEST_PARAMS_frozen.json`, whose values are listed in Table S1.10 and whose empirical confirmation against the committed outputs is described in the note to Section S1.

The absence of a per-run configuration record under `recon/` is a reproducibility weakness of these sweeps rather than of the model. It is the reason the seven values had to be recovered from run output and verified by replay, and it is the reason the parameter set is now committed as a file in its own right. The endogenous exit-capacity runs of Section S5 do not have this weakness, because each writes a `run_config.json`.

7.4 Recomputation of outcome classes

All reported regimes are recomputed from the continuous columns `final_exit_rate` and `max_punish` using the canonical classifier. No reported quantity reads a stored `regime` column. This rule is load-bearing rather than stylistic. The `regime` column committed alongside the `recon/` sweeps was written under an earlier classifier that applied a prevalence gate, and for the shock factorial that stored column disagrees with the canonical classifier on 71 percent of runs, labelling 2,880 runs MIXED that the canonical classifier labels CAPTURE. Stored regime columns are retained in the committed files as immutable simulation outputs, but they are not authoritative, and analysis recomputes classification from the continuous quantities every time.